

Particle Physics Homework 2

1) Rapidity

Another way of writing the boost transformation is writing velocity β in terms of rapidity η :

$$\begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cosh\eta & \sinh\eta & 0 & 0 \\ \sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

1.1). Show that $\eta = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$

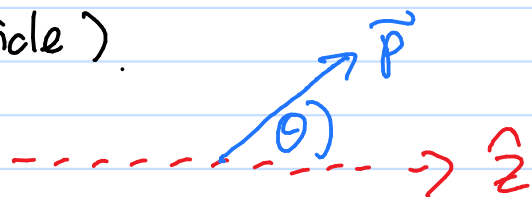
1.2). One of the advantages of using rapidity is the additional property. Show that

$$\Lambda(\eta_1) \Lambda(\eta_2) = \Lambda(\eta_1 + \eta_2)$$

1.3). In collider physics experiment, we usually deal with a very high energetic particle, i.e., its 4-momentum can be approximated as $p^\mu = (E, \vec{p}) \simeq (E, E\hat{p})$

(such that $p^\mu p_\mu \sim 0$). Consider a particle scattered off with a 3-momentum $\vec{p}$ and an angle θ with respect to $\hat{z}$ ($\hat{z}$ = direction of the incoming particle).

Show that the rapidity can



be approximated by pseudorapidity

$$\eta' = -\ln \left(\tan\left(\frac{\theta}{2}\right) \right)$$

2). 2-body decay of an unstable particle

Consider an unstable mother particle at rest with mass M . It eventually decay into 2 daughter particles with mass m_1 & m_2 respectively.

2.1) Show that the energies of daughter particles are

$$E_1 = \frac{1}{2M} (M^2 + m_1^2 - m_2^2)$$

$$E_2 = \frac{1}{2M} (M^2 + m_2^2 - m_1^2)$$

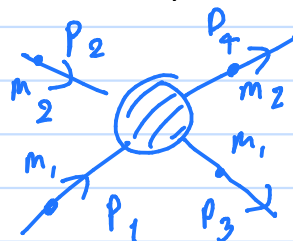
2.2) Show that the magnitude of momentum of daughter particles are

$$|\vec{p}| = \frac{1}{2M} \sqrt{(M^2 - (m_1 - m_2)^2)(M^2 - (m_1 + m_2)^2)}$$

2.3) Consider $E_1, E_2, |\vec{p}|$ in the limit that $m_1 = m_2$

3) Elastic Collision

Consider the case of elastic collision between 2 particles. The 4-momentum of 2 incident particles are p_1^μ & p_2^μ and the 4-momentum of scattered particles are p_3^μ & p_4^μ . The masses for 2 particles are m_1 & m_2 such that $p_1^\mu p_{1\mu} = p_3^\mu p_{3\mu} = m_1^2$ and $p_2^\mu p_{2\mu} = p_4^\mu p_{4\mu} = m_2^2$



There are 3 Lorentz invariants that are useful:

$$s = (p_1^\mu + p_2^\mu)(p_{1\mu} + p_{2\mu}) \equiv (p_1 + p_2)^2 = (p_3 + p_4)^2$$

$$t = (p_1^\mu - p_3^\mu)(p_{1\mu} - p_{3\mu}) \equiv (p_1 - p_3)^2 = (p_2 - p_4)^2$$

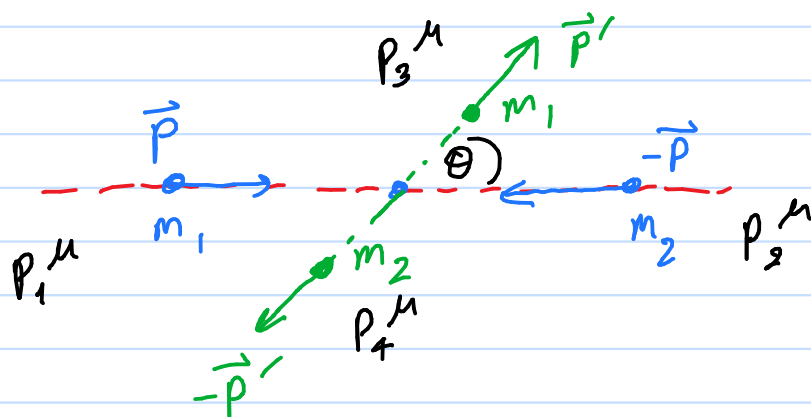
$$u = (p_1^\mu - p_4^\mu)(p_{1\mu} - p_{4\mu}) \equiv (p_1 - p_4)^2 = (p_2 - p_3)^2$$

These are called Mandelstam variables.

3.1) Show that $s + t + u = 2m_1^2 + 2m_2^2$

3.2) Let's calculate them in the center of momentum frame.

Consider $p_1^\mu = (E_1, \vec{p})$, $p_2^\mu = (E_2, -\vec{p})$



Show that $s = (E_1 + E_2)^2$. Therefore sometimes we call $\sqrt{s}$ = center of mass energy.

3.3) Show that in the center of momentum frame,

$$t = 2m_1^2 - 2E_1 E_3 + 2\vec{p} \cdot \vec{p}'$$

$$u = m_1^2 + m_2^2 - 2E_1 E_4 + 2\vec{p} \cdot \vec{p}'$$

3.4) Take the high energy limit ($m_1 \rightarrow 0$, $m_2 \rightarrow 0$)

Show that

$$t \approx -2E_1 E_3 (1 - \cos\theta)$$